

Simulation Scenario Report: ED Asthma Exacerbation – Siratech Downtime Protocol 1.

Section I: Scenario Demographics

- **Scenario Title:** Emergency Department Operational Readiness – Moderate Asthma Exacerbation during System Downtime.
- **Target Learning Group:**
 - ER Registration & HIM Staff ○ ER Nursing Staff (Triage and Treatment)
 - ER Physicians
 - Laboratory/Phlebotomy & Respiratory Therapy Staff ○ Radiology Technologists ○ Pharmacy Staff
- **Approximate Timing:**
 - **Set-Up:** 30 min
 - **Scenario:** 60 min (to account for manual downtime documentation) ○ **Debriefing:** 20 min

2. Section II:

Scenario Developers

- **Stakeholders:**
 - KFSHRC Hospital Operations (ED Leadership) ○ IT/Health IT (Downtime Support) ○ Emergency Nursing & Respiratory Services ○ Health Information Management (HIM)

3. Section III: Curriculum Integration

- **Educational Goal:** To validate the manual downtime workflow, clinical adherence to asthma protocols, and inter-departmental coordination when the Siratech (HIS) is unavailable.
- **CRM Objectives:**
 - **Manual Communication:** Demonstrate effective use of paper orders and verbal communication between ED modules and ancillary services.
 - **Patient Safety during Downtime:** Ensure 100% accuracy in manual patient identification and data tracking without electronic verification.
- **Operational Objectives:**
 - **Downtime Registration:** Validate staff proficiency in activating "downtime processes," including manual MRN assignment and handwritten ID armbands.
 - **Clinical Adherence:** Evaluate the speed of administering nebulization and steroids using downtime medication forms. ○ **Ancillary Coordination:** Confirm manual requisition process for blood work, respiratory swabs, and Chest X-rays.

4. Section IV: Scenario Script

- **Scenario Cast:**
 - **Patient:** Standardized Patient (25-year-old with moderate asthma).
 - **Confederates:** Actual ED staff performing their assigned real-world roles.
- **Required Equipment:**
 - **Downtime Kit:** Manual registration forms, paper lab/rad requisition slips, carbon-copy medication charts.

- **Clinical Tools:** Nebulizer machine, Albuterol/Ipratropium, IV Steroids, Phlebotomy kit, Viral Transport Media (VTM) for respiratory swab. ○ **Monitoring:** Continuous Pulse Oximetry and BP monitor.

Scenario Workflow Table:

Step/State	Action / System Trigger	Learner Actions (Downtime Protocol)	Auditor Checklist
Arrival (Visual Triage)	Patient arrives by private car walking to the ER entrance.	- Staff perform immediate Visual Triage. - Direct the patient/relative to the registration. - SARI score - Ask if need assistance.	- Assess speed of initial visual assessment. - Assess the acuricy of SARI score - Assess the encounter
Registration	System Down (Siratech).	Registrar uses manual downtime log • writes ID armband.	- Verify manual ID accuracy. - Assess the speed time for the registration process
Triage	Comprehensive Assessment.	- Nurse performs manual Triage; - assigns acuity (Level 3). Bed assignmet to the isolation	Check manual documentation of vitals.
In emergency room	Physician Assessment.	- Nurse recive the patient (intuduce self) • Nurse notify the doctor • Physician intrduce him self • writes Paper Orders for blood work Medication - Assess the physician encounter	• him/her Ensure "Time Ordered" is recorded.
Treatment	Med Administration.	Nurse/RT administers nebulization & steroids; Phlebotomist extracts blood.	Assess "Nurse-to-Patient" med delivery.
Diagnostics	Swab & X-ray.	Staff perform respiratory swab and transport patient to X-ray with paper slip.	Verify manual labeling of specimens.
Reassessment	Clinical Review.	Physician reassesses patient after treatment; documents on paper.	Evaluate clinical decision-making.
Discharge	Finalizing Care.	Physician completes manual discharge summary and paper prescriptions.	Ensure outpatient follow-up is noted.
Pharmacy	Med Collection.	Patient presents paper script to Pharmacy • for discharge medications.	Assess pharmacist's manual verification.

Post-Discharge

- Patient Experience calls the patient for OPD follow up and patient satisfaction survey
 - Assess the Quality of post discharge encounter
-

5. Section V: Patient Data and Baseline State

- **Clinical Vignette:** "A 25-year-old known asthmatic presents to the ER complaining of shortness of breath and chest tightness for 4 hours. The patient is speaking in partial sentences. Siratech is currently down for unscheduled maintenance."
- **Patient Profile:**
 - **Age:** 25 ◦ **PMH:** Bronchial Asthma
 - **Allergies:** None
- **Baseline State (Moderate Exacerbation):**
 - **O2 Sat:** 93% on RA (Triggers oxygen/nebulizer protocol) ◦ **RR:** 26/min ◦ **HR:** 110 bpm ◦ **Breath Sounds:** Bilateral expiratory wheezing.

6. Section VI: Supporting Documents & Results

- **Manual Requisitions:** ◦ **Lab:** CBC, Electrolytes, Arterial Blood Gas (if indicated). ◦ **Radiology:** Portable or Departmental Chest X-ray.
 - **Swab:** Respiratory Viral Panel (RVP).
- **Downtime Log:** Verification that the patient's manual MRN matches all handwritten labels.

7. Section VII: Debriefing Guide

- **Validation Questions:**
 - **Downtime Readiness:** How long did it take to transition from "System Down" awareness to using manual forms? ◦ **Communication:** Were the paper orders legible and delivered to the Lab/Radiology without delay?
 - **Patient Safety:** Was the manual ID armband checked before the administration of steroids and nebulization? ◦ **Workflow:** Did the lack of Siratech notifications (SMS or digital cues) lead to any "dead zones" in the patient's care?